# Diagnosing Human-to-Robot Transfer Failures with a Transfer Gap Budget in Simulation

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**Abstract:** A failed rollout does not show which stage of human-to-robot transfer failed. We introduce the *Transfer Gap Budget*, which replaces one pipeline component at a time with a declared oracle and measures the paired change in task success. In an injected-error study with 500 fresh seeds per arm, perception has the largest budget on both the UR5e and UR10e. A look-again correction, selected on development seeds, raises success by 26.8 and 25.8 pp, respectively, the largest observed gains among four corrections. Exploratory sensitivity runs change the largest budget: timing with speeds spanning the demonstrations (2.0–9.4$\times$ robot pace), calibration with reduced perception and doubled calibration error. Across three alternative error settings, budget selection beats look-again by 12–44 pp (95% intervals above 7 pp); it ties where perception dominates. On six demonstrations, diagnosis and a revised retreat raise paired success from 0–31/60 to 28–60/60 on unused seeds. Simulated depth perception meets pre-declared criteria on three manipulators. A G1 humanoid study finds 11/40 versus 31/40 placements for the same closed-loop controller across two runtime setups, with the cause unresolved. All robot evaluations are in simulation.

**Keywords:** human-to-robot transfer, failure diagnosis, oracle substitution

## 1 Introduction

Human-to-robot transfer requires motion estimation, retargeting, calibration, and executable timing. Success alone does not identify the faulty component across diverse human data [1]. The *Transfer Gap Budget* measures how much success each component's oracle recovers; gains can overlap. Development-set budgets select a correction for fresh evaluation.

## 2 Related Work

**Human-to-robot transfer.** H2RBench reports transfer gains by execution stage on a Franka arm [2]; we intervene on pipeline components. Human-as-Humanoid and HumanEgo learn from human video [3, 4]; hand-pose quality and motion mismatch affect cotraining [5, 6]. **Diagnosis and correction.** AHA, RoboFAC, and Guardian diagnose manipulation failures [7, 8, 9]. RHO and GaP search policy code and skill graphs [10, 11]. We use oracle gains to select from a fixed correction menu.

## 3 Method: the Transfer Gap Budget

**Paired interventions.** Let $\mathcal{K}$ contain perception, retargeting, tool-frame calibration, and execution timing. In the controlled-error study, *full* includes all four perturbations; $oracle_k$ removes only perturbation $k$, and *oracle-all* removes all four. Their ideal outputs are exact object pose, unperturbed IK, zero tool offset, and nominal timing. Each random stream is keyed by seed and component,

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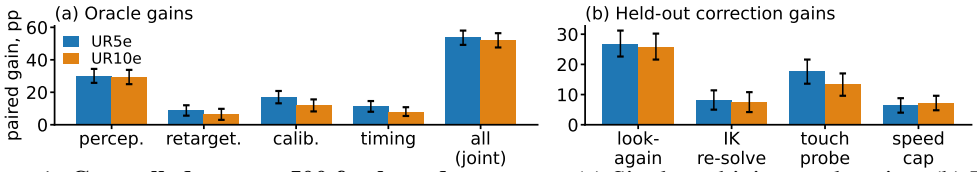



Figure 1: **Controlled errors: 500 fresh seeds per arm.** (a) Single and joint oracle gains. (b) Four correction gains; look-again selected on development seeds. Bars: paired bootstrap 95% intervals.

preserving initial scenes and unreplaced error draws across conditions. For binary success $s_c(i)$ on seed $i$ in condition $c$,

$$B_k = \frac{1}{n}\sum_{i=1}^{n}\big(s_{\text{oracle}_k}(i) - s_{\text{full}}(i)\big), \qquad \Delta_{\text{int}} = \sum_{k\in\mathcal{K}} B_k - B_{\text{all}}. \tag{1}$$

Here $B_{\text{all}}$ uses the joint oracle in the same paired difference. Budgets are signed net gains: an oracle can both rescue and lose trials. The residual shows whether summed single-stage gains exceed ($> 0$) or fall below ($< 0$) the joint gain; it does not allocate failures to interactions.

**Correction selection.** The menu pairs perception with look-again, retargeting with an IK re-solve, calibration with a modelled touch probe, and timing with a speed cap. We select $k^\star = \arg\max_k B_k$ on development seeds, breaking ties in pipeline order. Look-again and probing reduce assumed residual errors; their manoeuvres are not simulated.

**Statistics.** Paired percentile bootstraps resample seeds with all condition scores (95%). Demo intervals are descriptive for reused seeds, conditional on the six demonstrations for fresh seeds.

## 4 Experimental Setup

**Task.** MuJoCo simulates a fixed-base UR5e or UR10e with a Robotiq 2F-85 from MuJoCo Menagerie [12], a 4 cm block with $\pm$3 cm position jitter, and a tray with 14 cm inner width. A scripted IK controller uses simulated contact in a UR5e-tuned layout. Success requires a lift of at least 5 cm, release within 3 cm of the goal, and speed below 0.02 m/s for 0.5 s after release.

**A: controlled errors.** The injected magnitudes are assumptions. Both arms receive the same seed-dependent draws: Gaussian $xy$ perception error ($\sigma = 1.5$ cm), joint offsets ($\sigma = 0.5^\circ$ per joint), a uniformly directed tool-frame offset with magnitude $\mathcal{U}(0, 3)$ cm, and a timing speed-up of $\mathcal{U}(1.0, 4.5)$ relative to robot pace. The corrections reduce perception noise to 0.4 cm, joint-offset noise to $0.05^\circ$, or calibration noise to 2 mm per axis, or cap speed-up at $3\times$. Development seeds 0–59 select look-again on both arms. After freezing the protocol, we evaluate ten conditions on 500 fresh seeds (3000–3499) per arm: *full*, four single oracles, the joint oracle, and four corrections.

**E: exploratory sensitivity.** Fresh seeds (200 per arm unless stated) test A's oracles and selected corrections with errors scaled $0.5\times$ (500 seeds) or $2\times$, 3 or 5 cm blocks, $\pm$5 cm jitter, tray or block shifts of 8–10 cm, perception error scaled 0.1–$2\times$, cut to 25% alone or with doubled calibration error (500 seeds), matched to simulated depth-error measurements, and timing spanning demonstration speeds (500 seeds). Correction constants are unchanged.

**B: real-video motion.** Five pick-and-place cycles from two people in two freely licensed videos, plus one from a third person's 50 s recording, supply motion and timing. GEM-X [13] estimates hand trajectories; grasp and release frames are annotated manually. Reach and retreat are anchored to the simulated grasp and release points, and the carry path is rotated and scaled to connect them; the scene is not reconstructed. Perception and calibration errors remain injected; no object detector is used, and retargeting is excluded from the real-video oracle analysis. Each demo–arm pair uses 60 seeds from the revision period (1000–1059) and 60 unused seeds (5000–5059). We report leg-wise path RMSE against the retargeted reference.

**C: depth perception.** A geometric estimator fits the table plane, clusters points above it, and selects a cube and tray; its position, yaw, and size drive the controller. We test 5, 6, and 7 cm cubes on

the two arms and a third manipulator (UR10e with a Robotiq 2F-140), with 40 seeds per size: 120 episodes per condition. **D: humanoid.** A Unitree G1 with Dex3 hands uses SONIC whole-body control [14]. We score grasp (5 cm lift with contact) and placement within 5 cm of the goal; we do not estimate stage budgets here.

# 5 Results

## 5.1 Controlled errors on two arms

Uncorrected success is 232/500 on the UR5e and 240/500 on the UR10e. Removing all errors recovers 53.6 and 52.0 pp. Perception has the largest budget: 30.0 [25.8, 34.4] pp on the UR5e and 29.4 [25.0, 33.8] pp on the UR10e. Calibration follows (17.0 and 12.0 pp), then timing (11.2 and 8.0 pp) and retargeting (8.8 and 6.4 pp); all single-stage intervals exclude zero (Fig. 1a).

The interaction residual is 13.4 [6.2, 20.8] pp on the UR5e and 3.8 [-2.6, 10.4] pp on the UR10e.

Look-again, selected on development trials, raises success from 232 to 366/500 (UR5e) and 240 to 369/500 (UR10e); the paired gains are 26.8 [22.6, 31.2] pp and 25.8 [21.6, 30.2] pp, respectively, the largest observed. Probing gives 17.6 and 13.4 pp; IK re-solving and the speed cap 6–8 pp (Fig. 1b).

## 5.2 Sensitivity to assumptions, layout and the budget rule

Perception has the largest point-estimate budget at nominal and doubled errors, with 3 or 5 cm blocks, $\pm$5 cm jitter (ten cells), and moved tray or block (six cells). At $0.5\times$ errors (484 and 485 of 500 uncorrected) it still leads, by 0.8 pp [0.2, 1.6] over calibration on both arms. Varying only perception error (Fig. 2), perception leads from the assumed 1.5 cm upward; at $0.75\times$ and below calibration and timing take over.

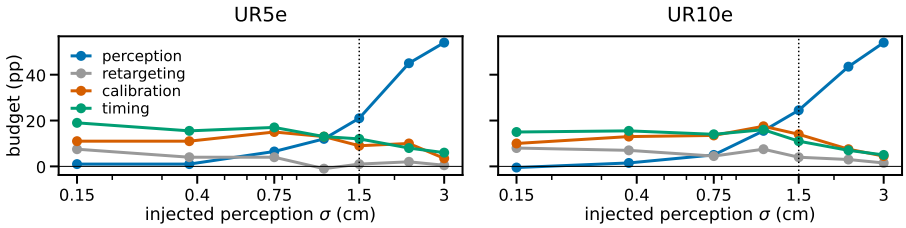



Figure 2: Point-estimate budgets against perception error (200 seeds per point); dotted: assumed value.

Tying magnitudes to measurements changes the ranking. The simulated depth estimator's block-position error is 0.34 mm RMS per axis with exact camera calibration, and 7.1 and 14.2 mm with $0.5°$/5 mm and $1°$/10 mm calibration error. Injected as the perception error, these give perception budgets of $-1.0$ and 2.0 pp, 10.0 and 9.5 pp (timing leads at both), and 28.5 and 31.0 pp (perception leads). The six demonstrations run at 2.0–9.4$\times$ robot pace, extending beyond the assumed 1–4.5$\times$. With speed drawn uniformly from $\mathcal{U}(2.01, 9.38)$ (72 and 87 of 500 uncorrected), timing has the largest budget, 50.2 and 44.2 pp, 38.6 [32.6, 44.2] and 35.8 [30.0, 41.4] pp above the next stage. Calibration and retargeting magnitudes remain assumed.

Each 500-seed setting uses 100 development and 400 evaluation seeds. With perception error cut to 25%, the rule picks timing: gains 14.0 and 13.0 pp versus 2.2 and 1.0 pp for look-again, a difference of +11.8 [7.5, 16.0] and +12.0 [7.8, 16.2] pp. It trails hindsight-best calibration by 1.2 and 1.5 pp (difference intervals include zero). With calibration error also doubled, it picks calibration: gains 43.8 and 44.5 pp versus about zero for look-again, differences +43.8 [39.0, 48.5] and +44.0 [39.0, 49.0] pp. With the demonstration-based timing range, it picks timing: gains 42.5 and 44.0 pp, exceeding look-again by +31.8 [25.0, 38.2] and +35.2 [28.7, 41.8] pp. The latter two match the hindsight best. Where perception dominates, the rule picks look-again and ties.

### 5.3 Real-video demonstrations

Table 1: Successes of 60 paired seeds, uncorrected → revised stack; left seeds were reused in development, right seeds not. P1–P3: demonstrators.

| | Seeds 1000–1059 | | Unused seeds | |
|---|---|---|---|---|
| Demonstration | UR5e | UR10e | UR5e | UR10e |
| P1 notebook | 0→ 57 | 0→ 60 | 0→ 60 | 0→ 60 |
| P1 markers | 0→ 51 | 0→ 25 | 0→ 49 | 0→ 28 |
| P1 orange marker | 0→ 46 | 0→ 58 | 0→ 51 | 0→ 59 |
| P1 notepad | 0→ 58 | 0→ 58 | 0→ 60 | 0→ 60 |
| P2 box (own stack) | 11→ 60 | 4→ 55 | 23→ 60 | 9→ 49 |
| P2 box (P1 stack) | 11→ 60 | 4→ 50 | 23→ 60 | 9→ 52 |
| P3 recording (P1 stack) | 35→ 59 | 30→ 56 | 31→ 57 | 21→ 53 |

The notebook demo fails uncorrected on both arms at about $6\times$ robot pace. Sequential diagnosis selects a $3\times$ speed cap, look-again, calibration probing, then a tighter $2\times$ cap; with the revised retreat the stack reaches 57/60 and 60/60 (Table 1), and RMSE to the retargeted reference falls from 13.9 to 4.9 cm (UR5e) and 9.1 to 2.9 cm (UR10e).

The original stack scored 0/60 on all 3 additional P1 cycles, whose copied retreat paths intersect the tray, and cut P2 success on the UR5e from 11/60 to 0/60. We therefore added a 5 cm lift before retreating along the human path (selected on development seeds 0–19; path-RMSE cost 2.7–4.1 cm). The revised P1 stack transfers to P2 and P3 without retuning; P1 markers remain difficult on the UR10e (25/60). Seeds 1000–1059 informed revision; those columns are exploratory. On unused seeds 5000–5059, every cell improves: the revised stack reaches 28–60/60 against at most 31/60 uncorrected, all paired intervals exclude zero, and P1 markers on the UR10e reach 28/60.

### 5.4 Depth perception and humanoid runtime sensitivity

All 30/30 pre-declared acceptance cells reach the 90% point-estimate threshold (113–120 of 120 episodes), covering yaw, distractors, depth noise, occlusion with a retry, and combined stressors on three manipulators. Report-only stress tests expose limits: $2^\circ$/20 mm calibration error gives 25/120 on the UR5e, and occlusion without retry 0–16/120. For the notebook path at robot pace, depth versus ground-truth success is 109 versus 108/120 (UR5e), 112 versus 111/120 (UR10e), and 120 versus 118/120 (third arm): 0.8–1.7 pp higher, with path RMSE within 0.01 cm. All three meet the point-estimate margins of 10 pp success loss and 0.5 cm added RMSE (no inferential non-inferiority test).

**G1 humanoid.** With IK-planned targets, the same frozen closed-loop controller places 11/40 objects in a volume-based installation and 31/40 in a packaged container (same GPU, seeds 1200–1239; discordant placements 20 to 0, exact McNemar $p = 1.9 \times 10^{-6}$). Code and environment checks did not identify the cause; a four-seed MuJoCo 3.15.0 rebuild (0/4 lifts versus 4/4 on 3.14.0) also changed `filelock`.

## 6 Limitations and Conclusion

The study uses one simulated task, scripted controllers, and a UR5e-tuned layout. Error distributions and correction models remain assumed; depth-error magnitudes are measured in simulation and timing ranges from demonstrations. Layout shifts leave the table and task fixed. Six demonstrations from three people and unreconstructed scenes limit generalisation; look-again and probing cost no simulated time. Perception leads under the assumed model; timing leads with demonstration-based speeds, and budget selection follows. Real-robot performance is untested. Generative AI was used throughout manuscript and code drafting; values were checked against saved results.

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